Exercise Sheet 28

The invariant form is normalised so that long roots have squared length 2. Verma modules are infinite-dimensional algebraic modules. Exercises 28.1–28.3 compute central characters. Exercises 28.4 and 28.5 compare split and nonsplit extensions. Exercise 28.6 uses central characters to find isotypic components.

Exercise 28.1 (Rank-one Casimir). Let $C = \frac{1}{2}h^2 + ef + fe \in U(\mathfrak{sl}_2)$.

(i) Show directly that $[e, C] = 0$ in $U(\mathfrak{sl}_2)$.

(ii) In the basis $v_j = f^j v_0$ $(j = 0, 1, 2)$ of V_2, write the matrices $D(h)$, $D(e)$ and $D(f)$ of h, e and f. Compute $D(C)$.

(iii) This part returns to Exercise 16.1 in the notation of Lecture 28. Find the Casimir element C_K of the Killing form and its eigenvalue on V_m. Check the value on the adjoint module.

(iv) Prove that C acts on $M(\lambda)$ and $M(\mu)$ by the same scalar if and only if $\mu \in \{\lambda, -\lambda - 2\}$.

Exercise 28.2 (The shifted orbit of 0 in A_2). For $\mathfrak{sl}_3$ write weights by Dynkin labels $(a, b) = a\omega_1 + b\omega_2$. Then $\alpha_1 = (2, -1)$, $\alpha_2 = (-1, 2)$ and $\rho = (1, 1)$.

(i) Using $s_1(a, b) = (-a, a + b)$ and $s_2(a, b) = (a + b, -b)$, compute the six weights $w \cdot 0 = w\rho - \rho$. Compare them with Figure 28.1.

(ii) Show that $\lambda = (a, b)$ satisfies $(\lambda, \lambda + 2\rho) = \frac{2}{3}(a^2 + ab + b^2) + 2(a + b)$. Check that all six weights give 0.

(iii) Let $e_i \in \mathfrak{g}_{\alpha_i}$ and $f_i \in \mathfrak{g}_{-\alpha_i}$ be simple-root vectors with $[e_i, f_i] = h_i$. In $M(0)$, with highest-weight vector v_0, show that $f_1 v_0$ and $f_2 v_0$ are killed by $\mathfrak{n}_+$.

(iv) Which of the six Verma modules $M(w \cdot 0)$ has a nonzero finite-dimensional quotient?

Exercise 28.3 (A cubic central character in A_2). Identify $\mathfrak{h}^*$ for $\mathfrak{sl}_3$ with $\{\nu \in \mathbb{C}^3 : \nu_1 + \nu_2 + \nu_3 = 0\}$, where $\nu(\mathrm{diag}(x_1, x_2, x_3)) = \sum_i \nu_i x_i$. Then $W = S_3$ permutes coordinates. Let $p_2 = \sum_i \nu_i^2$ and $p_3 = \sum_i \nu_i^3$.

(i) Show that $\gamma(\Omega) = p_2 - 2$ for the trace form.

(ii) Since p_3 is W-invariant, Harish-Chandra's theorem gives $z_3 \in Z$ with $\gamma(z_3) = p_3$. Compute χ_{ω_1} and χ_{ω_2} on Ω and z_3.

(iii) Let w_0 be the longest element of W. For $\lambda \in P_+$ and $\lambda^* = -w_0\lambda$, show that $\chi_{\lambda^*}(\Omega) = \chi_\lambda(\Omega)$ and $\chi_{\lambda^*}(z_3) = -\chi_\lambda(z_3)$.

Exercise 28.4 (Different central characters split). Let $\mathfrak{g}$ be semisimple and $0 \to A \to E \to B \to 0$ an exact sequence of $\mathfrak{g}$-modules, of any dimension. Suppose Z acts on A by a character χ and on B by $\chi' \neq \chi$.

(i) Prove that the sequence splits.

(ii) Deduce, without Weyl's theorem, that an $\mathfrak{sl}_2$-module E with a submodule $A \cong V_n$ such that $E/A \cong V_m$, $m \neq n$, is isomorphic to $V_n \oplus V_m$.

Exercise 28.5 (A nonsplit extension with one central character). In the $\mathfrak{sl}_2$ Verma module $M(0)$ use the basis $v_j = f^j v_0$.

(i) Show that $N = \operatorname{span}\{v_j : j \geq 1\}$ is a submodule isomorphic to $M(-2)$ and that $M(0)/N \cong \mathbb{C}$.

(ii) Show that N has no complementary submodule.

(iii) Compare the central characters and characters of $M(0)$ and $M(-2) \oplus \mathbb{C}$. Why do Weyl's theorem and Exercise 28.4 not apply?

Exercise 28.6 (Central characters and isotypic components). Let $\mathfrak{g}$ be semisimple, V a finite-dimensional $\mathfrak{g}$-module and $\lambda \in P_+$. Put

$$V^{\chi_\lambda} = \{v \in V : zv = \chi_\lambda(z)v \text{ for all } z \in Z\}.$$

(i) Using the isotypic decomposition of Lecture 14 and the separation theorem of Lecture 28, prove that V^{χ_λ} is the λ-isotypic component of V.

(ii) For $\mathfrak{sl}_3$ and $V = \mathbb{C}^3 \oplus (\mathbb{C}^3)^*$, find the $\frac{8}{3}$-eigenspace of Ω and $V^{\chi_{\omega_1}}$.